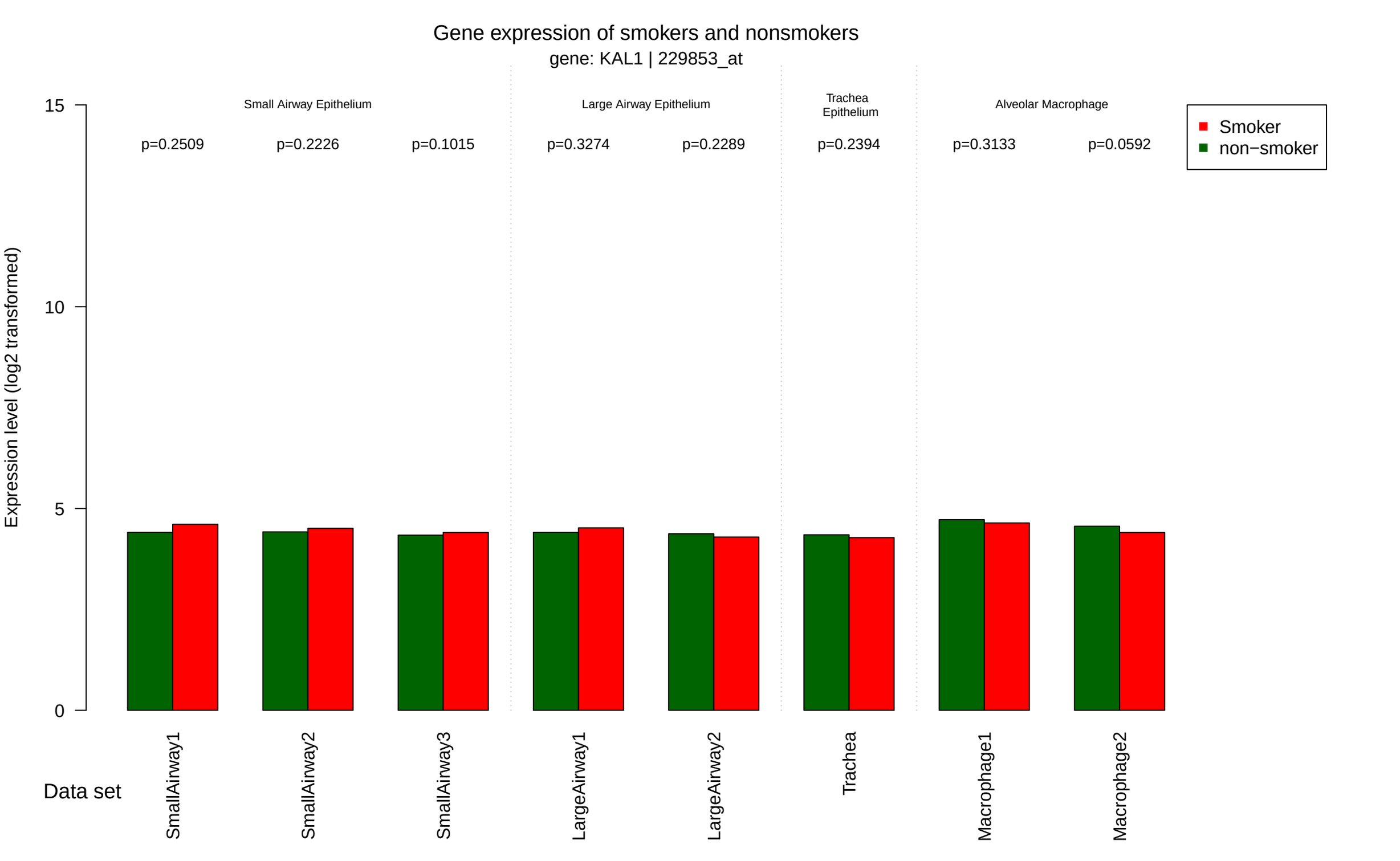
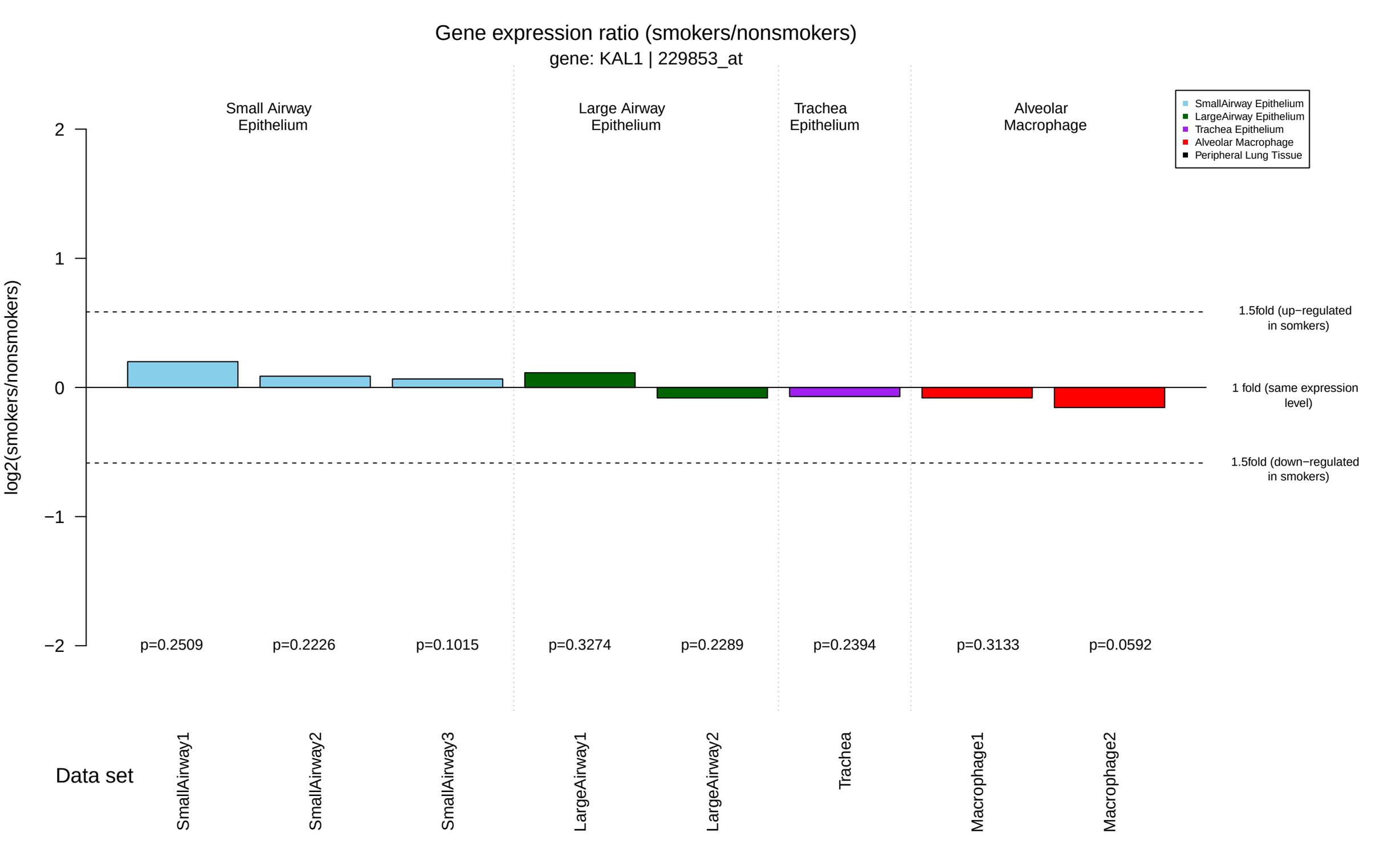
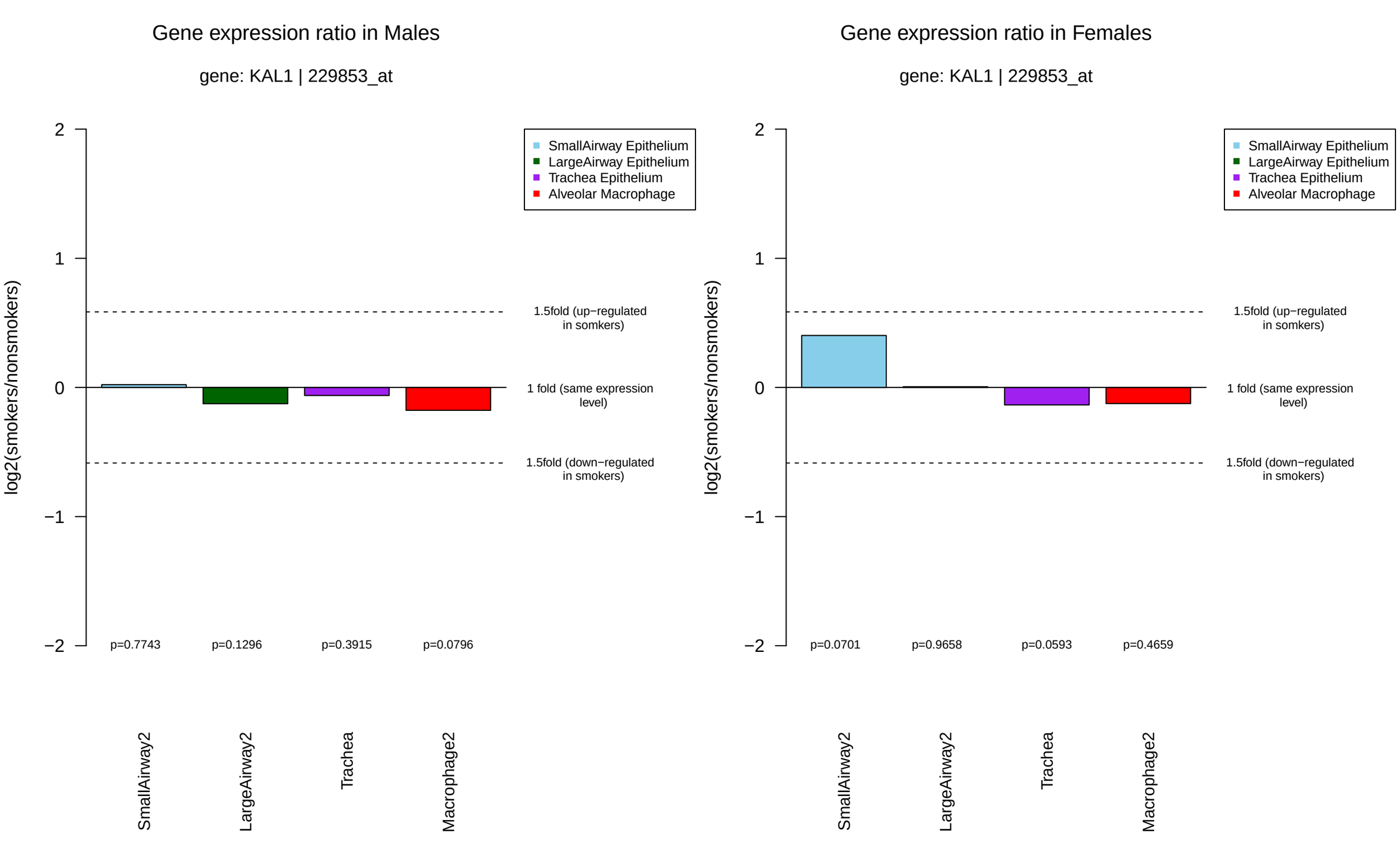
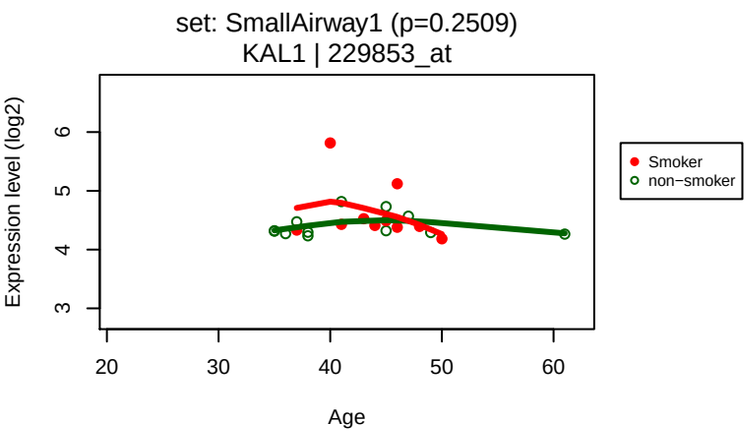
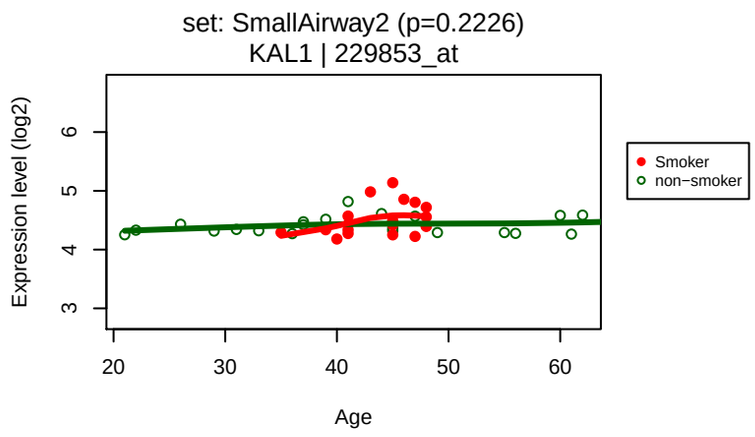
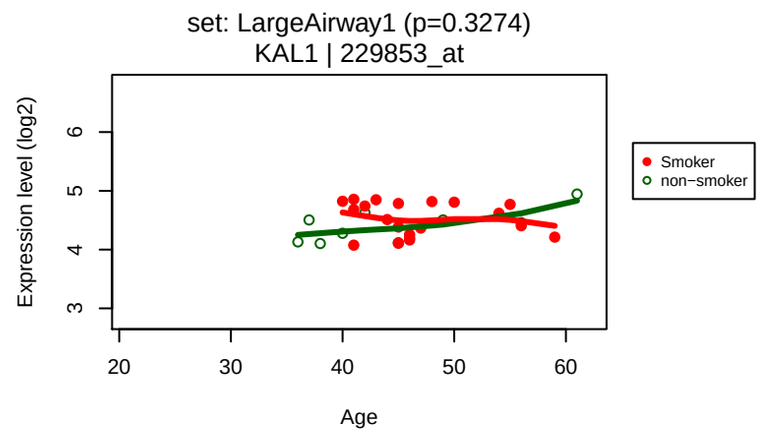

Gene expression of smokers and nonsmokers
gene: KAL1 | 229853_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2509
p=0.2226
p=0.1015
p=0.3274
p=0.2289
p=0.2394
p=0.3133
p=0.0592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

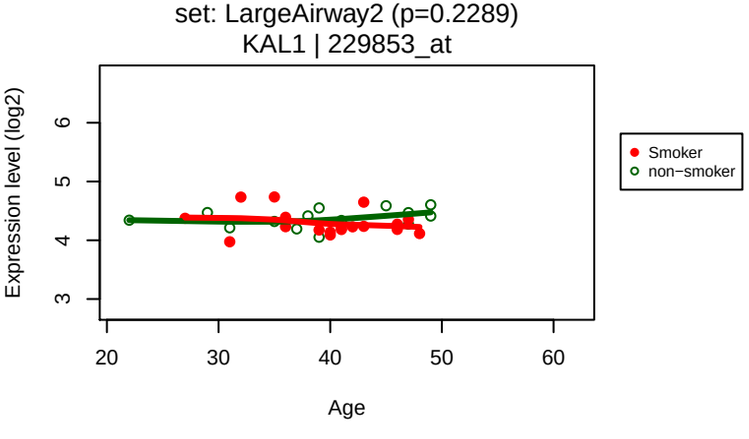

Gene expression ratio (smokers/nonsmokers)
gene: KAL1 | 229853_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2509
p=0.2226
p=0.1015
p=0.3274
p=0.2289
p=0.2394
p=0.3133
p=0.0592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

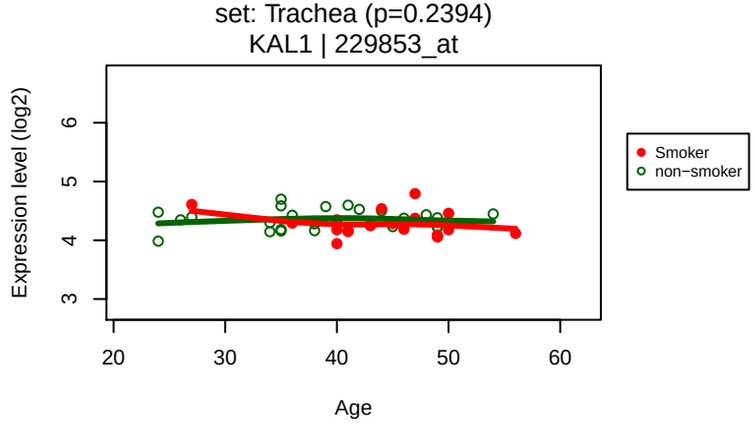

Gene expression ratio in Males
gene: KAL1 | 229853_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7743
p=0.1296
p=0.3915
p=0.0796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KAL1 | 229853_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0701
p=0.9658
p=0.0593
p=0.4659
SmallAirway2
LargeAirway2
Trachea
Macrophage2

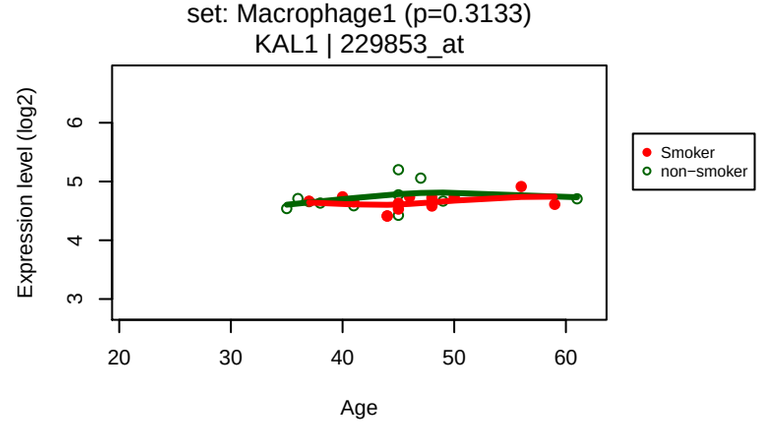

set: SmallAirway1 (p=0.2509)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker

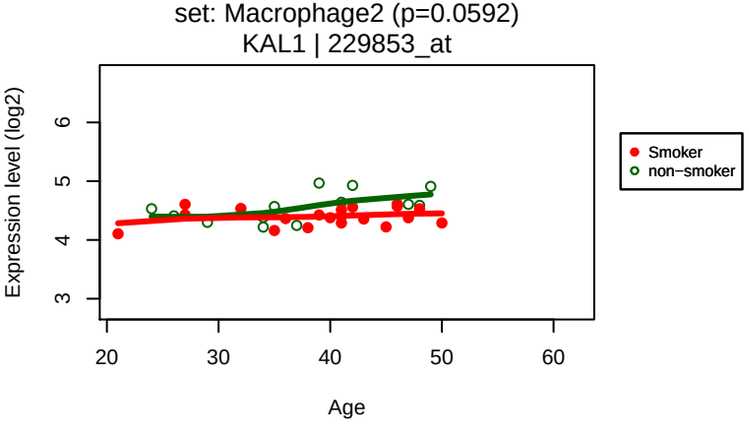

set: SmallAirway2 (p=0.2226)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker

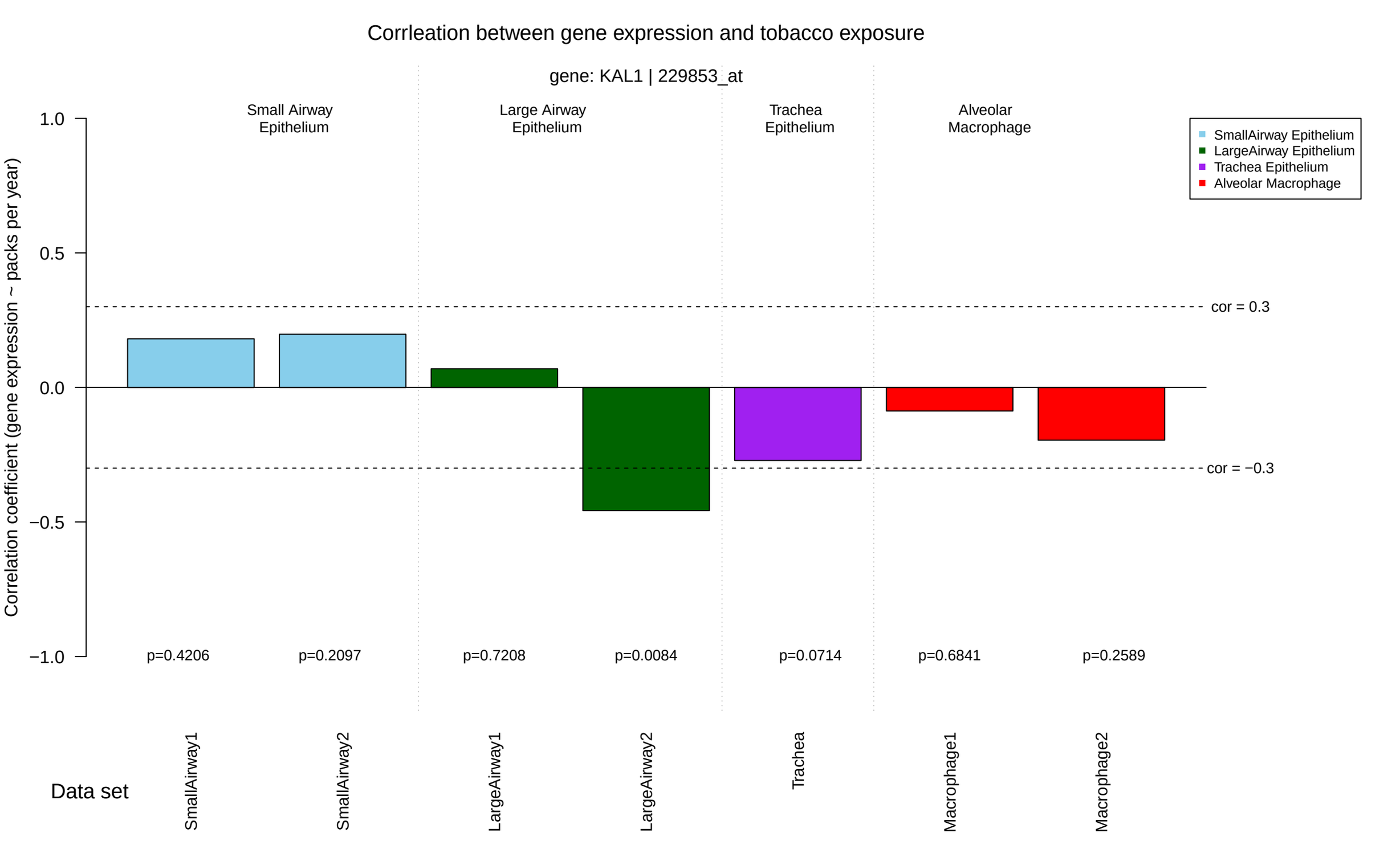

set: LargeAirway1 (p=0.3274)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2289)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2394)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3133)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0592)
KAL1 | 229853_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KAL1 | 229853_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4206
p=0.2097
p=0.7208
p=0.0084
p=0.0714
p=0.6841
p=0.2589
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2